



Proteomics assessment of CSF and blood plasma of the PPMI cohort using Olink Explore 1536 (Part 2)

This document was prepared by **Abbvie, Olink Support, Olink Proteomics** in consultation with the **AMP PD Proteomics Working Group**.

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PPMI Project ID: 222

Summary

Proteomics analysis was conducted on cerebrospinal fluid and blood plasma of both Parkinson's Disease patients and healthy participants in the PPMI cohort. Analysis was conducted using Olink Explore for 225 participants run on 666 CSF samples & 898 matched plasma samples, which uses two DNA tagged antibodies per analyte and amplification in a Proximity Extension Assay (PEA) for measuring protein levels. The method requires data to be converted into Normalized protein expression (NPX) values to be used in downstream analysis. NPX data is then intensity normalized and undergoes extensive quality control using the different assay controls before proceeding with analysis.

Use Note: Project 196 and 222 both contain NPX values generated via Olink Explore 1536; bridging samples were included in both projects to allow the files to be combined for analysis purposes. However, caution should be exercised in analyzing these projects together, since Olink recommends the use of formal bridging protocol for such situations. PPMI has provided a bridged file for these analyses under Project 9000 for users' convenience. Investigators should carefully review the information provided on the bridging method and confirm that the analytical methods applied meet their research objectives. Further, investigators interested in combining PPMI projects with other Olink results should strongly consider reaching out to Olink Biostatistics for guidance.





Abbreviations

CoA	Certificate of Analysis
CSF	Cerebrospinal Fluid
fixSD	A predefined standard deviation
LOD	Limit of detection
MAD	Median absolute deviation
NPX	Normalized protein expression
PC	Plate control
PEA	Proximity Extension Assay
NGS	Next Generation Sequencing

Acknowledgements

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Associated Files

Normalized Protein Expression Files

“PPMI_Project_222_{Sample Tissue Type}_{Assay Type}_NPX_{download date ddmmmyyyy}.csv”

Sample Tissue Types: “CSF”, “Plasma”

Assay Types: “CARDIO”, “NEURO”, “ONC”, “INF”

Data is presented as NPX (Normalized Protein eXpression) values. NPX is Olink's relative protein quantification unit on log2 scale. NPX values from Olink® Explore 384 and Olink® Explore 1536 (a combination of 4 separate Olink® Explore 384 panels) are calculated from the number of matched counts, using NGS (Next Generation Sequencing) as readout. The NPX values are presented in a separate results file delivered in the MyData cloud. Data values for measurements below limit of detection (LOD) are reported for all samples.





Column headers glossary

Term	Description	Utility in Analysis
PATNO	Project assigned patient ID	Project specific ID
EVENT_ID	Visit for sample	Timepoint of collection
INDEX	Number corresponding to the index sequence used to label each sample in Olink workflow	Minimal, as the sample is already identified by the sampleID at the NPX file level
OLINKID	Assay (protein being evaluated in a particular line); unique identifier on the vendor side	Minimal, mostly for software recognition of each each
UNIPROT	Six-digit alphanumeric code derived from UniProt data base. Assays that lack a UniProt ID well have distinct syntax; Refer to https://www.uniprot.org/	This enables rapid lookup of significant results obtained in analysis. Naturally, it's difficult to have a strong command of 1400+ protein, Uniprot can help identify some biology Olink may have discovered
ASSAY	Common protein name of the protein being assayed by Olink	In addition to sampleID, this is the main descriptor of the dataset
MISSINGFREQ	For a given assay, the fraction (out of 1) of data beneath the limit of detection (see below)	Helps in identifying low expression proteins.
PANEL	Which panel a given assay is on	Useful to know if a large portion of proteins of interest are in a specific panel
PANEL_LOT_NR	Panel lot number	
PLATEID	Identifier of which plate the sample and assay was run on	
QC_WARNING	Binary; identifies samples that have passed Olink QC	Olink recommends excluding the data with QC warning in your data analysis





LOD	Limit of detection; result of three standard deviations beneath the mean NPX value for a given assay; in Olink Explore, LOD is calculated for each plate individually	Olink recommends that assays with a large proportion of samples below LOD is excluded from the analysis. The limit for exclusion should be decided on a study basis and consider design including sample size and experimental variables. Suitable exclusion limits may be in the range of less than 25-50% of samples above LOD. For more information refer to: https://www.olink.com/question/how-is-the-limit-of-detection-lod-estimated-and-handled/
NPX	Normalized protein expression; calculated for each assay based on internal and external controls	Log 2 scale; statistically ready for simpler approaches like T-test/ANOVA as well as higher dimensional approaches like principal component analysis, neighborhood embedding, etc

List of Documents:

PPMI_Project_222_CSF_Cardio_NPX_{download date ddmmyyyy}.csv
PPMI_Project_222_CSF_INF_{download date ddmmyyyy}.csv
PPMI_Project_222_CSF_NEU_{download date ddmmyyyy}.csv
PPMI_Project_222_CSF_ONC_NPX_{download date ddmmyyyy}.csv
PPMI_Project_222_Plasma_CARDIO_NPX_{download date ddmmyyyy}.csv
PPMI_Project_222_Plasma_INF_NPX_{download date ddmmyyyy}.csv
PPMI_Project_222_Plasma_NEURO_NPX_{download date ddmmyyyy}.csv
PPMI_Project_222_Plasma_ONC_NPX_{download date ddmmyyyy}.csv





Method

1. Assay controls

We have an extensive quality control procedure that we follow when running an assay. This allows for full visibility and control over the technical performance of the assay at each step and ensures that reliable data are generated with customers' samples.

1.1 Internal controls

Three internal controls are added to each sample to monitor the quality of assay performance, as well as the quality of individual samples:

- An Incubation Control, which is a PEA assay for spiked-in non-human proteins.
- An Extension Control, which is an antibody molecule with both DNA-oligos attached and therefore always in proximity.
- An Amplification Control, which is a synthetic double-stranded amplicon.

The extension control is used to calculate the NPX, and the other two are used for quality control of the assay.

1.2 External control

We also include external control samples on each plate to normalize data and also to monitor the assay's performance:

- A Sample Control of pooled plasma to estimate intra- and inter-assay precision.
- A Negative Control to measure background levels, which is used to estimate LOD for all assays.
- A Plate Control of pooled plasma to calculate the NPX and normalize signal levels between different plates for each assay.

2. Converting counts to NPX

The Explore system's raw data output are NGS counts, where each combination of an assay and sample is given an integer value based on the number of DNA copies detected. These raw data counts are converted into NPX values for use in downstream statistical analysis.

2.1 NPX generation and Normalization

The NPX values are calculated in two main steps and then intensity normalized by performing between-plate-normalization. First, the assay counts of a sample are divided by those of the extension control for that sample and block, which then undergoes log₂ transformation to normalize the data:

Steps in the NPX generation described in equation form, where "i" refers to a specific assay, j refers to a sample, and ExtNPX defines an extension normalized NPX value.

1. $ExtNPX_{i,j} = \log_2(counts(sample_jAssay_i)/counts(ExtCtrl_j))$
 - a. Relate counts to a known standard (Extension control)





- b. For all assays and all samples, including negative controls, Plate Controls, and Sample controls.
- c. Log₂ transformation gives more normally distributed data

The result is a scale that has increasing sample values against increasing protein concentration for each assay. The median of the Plate Control is then subtracted from the normalized data:

2. $\text{NPX}_{i,j} = \text{ExtNPX}_{i,j} - \text{median}(\text{ExtNPX}(\text{Plate Controls}_i))$
 - a. Perform plate standardization
 - b. For all assays and per plate of samples

Intensity normalization of data is the default for randomized studies on multiple sample plates. In this setting, the median for a random selection of samples is more stable across plates than the three PC's on each plate. Intensity normalization sets the median level of all assays to the same value for all plates:

3. $\text{NPX_Intnorm}_{i,j} = (\text{NPX}_{i,j} - \text{plate median}(\text{NPX}_i)) + \text{global median}(\text{NPX}_i)$
 - a. Between plate normalization (Optional but the default for multi-plate projects)
 - b. For each assay, for all plates in the project.

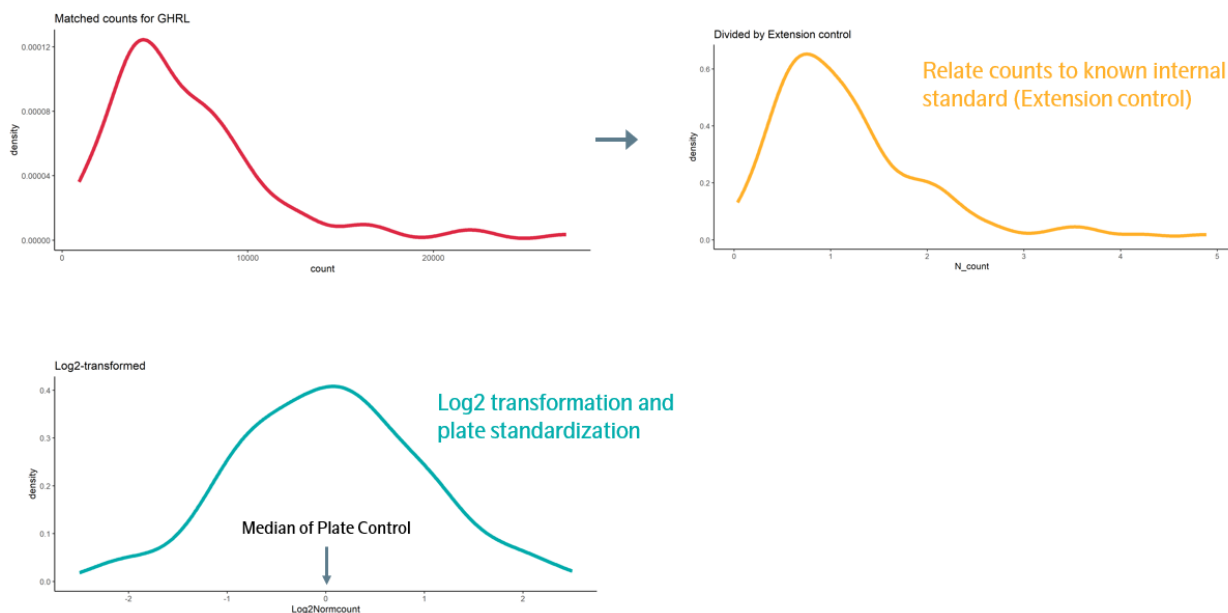


Figure 1: Summary of the three steps involved in NPX generation and data normalization. Step 1 is illustrated by the red graph, step 2 by the yellow graph, and step 3 (data normalization) by the turquoise graph. GHRL refers to appetite-regulating hormone (Uniprot ID: Q9UBU3).





3. Quality control

3.1 Sample QC

Each sample in a block is given the status *pass* or *warning*. For a complete Explore 1536 run, this means that each sample is evaluated 16 times. There are three sample QC criteria per block. Exceeding specifications in any one of them will result in a QC warning status for that sample. The three criteria are listed below:

1. Incubation control deviation from the median may not exceed 0.3 NPX.
2. Amplification control deviation from the median may not exceed 0.3 NPX.
3. Average counts for a sample may not fall below 500 counts.

Data from samples with a warning should be treated with caution.

3.2 Run QC

In addition to specific samples that exceed specifications, the entire block can be considered failed if it meets criteria 1 and 2 below. For practical reasons, this will most likely result in the re-run of a panel. If criteria 3 below is met the panel is failed:

1. The number of samples with QC warning exceeds 16/88 or 1/6.
2. The MAD for incubation or amplification control across all samples exceeds 0.3.
3. The Plate Control and Negative Control exceeds the criteria specified in section *Control strip QC*.

3.3 Control strip QC

The Plate Controls and negative controls have specific QC criteria. The median of the triplicates may not deviate more than 3 standard deviations from predefined tolerance levels for more than 10% of the assays on average for a panel. For the negative control, only positive deviations are considered.

4. LOD and CV calculation

LOD is defined as being three standard deviations above the median NPX of negative controls. The median is set using all samples annotated as negative controls per plate. A predefined standard deviation is used (fixSD). Detectability is calculated per assay and plate and is defined by the percentage of samples above the LOD threshold. The overall detectability of the project is generated and reported in the CoA:

The CV is calculated per assay (i) using the assumption of a log-normal distribution. The average CV is then calculated across panels and included in the CoA output.

$$CV_i = 100 * \sqrt{e^{Sln_i^2} - 1}, \text{ where } Sln_i = \ln(2) * SD_i$$





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References

Explore 384 Cardiometabolic (CARDIO) panel: <https://www.olink.com/products/olink-explore/olink-explore-384-cardiometabolic/>

Explore 384 Inflammation (INF) panel: <https://www.olink.com/products/olink-explore/olink-explore-384-inflammation/>

Explore 384 Neurology (NEU) panel: <https://www.olink.com/products/olink-explore/olink-explore-384-neurology/>

Explore 384 Oncology (ONC) panel: <https://www.olink.com/products/olink-explore/olink-explore-384-oncology/>

All four of the above panels together is the Olink Explore 1536 panel:
<https://www.olink.com/products/olink-explore/>

Data normalization and standardization
https://www.olink.com/content/uploads/2018/18/Data-normalization-and-standardization_v1.0.pdf

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